Student Name: Mutuku Geoffrey Mulu
Admission Number: F18/144332/2022
Level: 5th Year Mechanical Engineering
Project Title: Development of a RAG-Enabled Local Large Language Model for Autonomous Mechanical Material Selection

1 Introduction

Material selection is a core decision-making task in mechanical design that directly impacts a component's cost, quality, and structural integrity. Traditionally, selecting the optimal material requires deep technical expertise to analyze complex boundary constraints and environmental factors. Translating non-technical functional requirements into precise material specifications remains a significant bottleneck for rapid prototyping. While artificial intelligence offers potential solutions, applying standard Large Language Models (LLMs) directly to critical manufacturing tasks introduces the severe risk of "hallucination," where the system generates plausible but factually incorrect material properties.

2 Project Objective

This project aims to strictly design and develop a standalone, natural-language-driven material selection software tool. The system will allow users to input functional, non-technical descriptions of a desired component (e.g., "I want to create a spoiler for a car" or "I need to make blades for a wind turbine"). The tool will process this prompt and output the most suitable material recommendations, alongside required specifications and structural constraints. The project focuses entirely on the material selection algorithm. Any specific physical components evaluated by the software will serve purely as digital test cases to validate the algorithm's accuracy, not as physical manufacturing deliverables.

3 Proposed Methodology

To eliminate AI hallucination and ensure absolute engineering accuracy, the system will utilize a Retrieval-Augmented Generation (RAG) architecture. Instead of relying on the LLM's internal memory to guess parameters, the RAG framework connects the AI to a verified external database of material data sheets and engineering formulas in real-time. The system will be designed to run as a standalone, locally hosted tool to ensure continuous functionality without requiring internet access.

4 Expected Deliverables

The final deliverables will be the functional material selection software tool and the accompanying technical report. The scope is strictly contained to the development of the Natural Language Processing (NLP) pipeline, the RAG database integration, and the frontend user interface.